The Variance Risk Premium as a Return Predictor

Replication of Bollerslev, Tauchen & Zhou (2009)

Expected Stock Returns and Variance Risk Premia, Review of Financial Studies 22(11), 2009

- VRP = IV − RV = $VIX^2/12 - \Sigma r^2_d$ (variance risk premium construction)
- Epstein–Zin recursive preferences: why volatility-of-volatility creates return predictability
- Horizon regressions h = 1–24 months with Newey–West standard errors
- Sign reversal in 2008–2026: the paper's result does not survive out-of-sample

Course Roadmap

1 **Theory**
The VRP definition and its equilibrium foundations (Epstein–Zin)

2 **Signal Construction**
Computing IV, RV, and VRP from daily SPX and VIX data

3 **Statistical Validation**
Horizon regressions, sign reversal, and outlier diagnostics

4 **Position Sizing**
Translating VRP into binary and scaled trading signals

5 **Backtest**
Cumulative returns, Sharpe ratio, and drawdown analysis

6 **Paper Comparison**
Table 2 replication: our slopes vs. the paper's slopes

7 **Robustness Checks**
Ex-Oct-2008, volatility form, and HAR-RV expected VRP

8 **Honest Assessment**
What the data supports and what it does not

1 Theory

The VRP and Its Equilibrium Foundations

What Is the Variance Risk Premium?

The VRP: Plain-Language Definition

- The VIX is the market's price of variance insurance
- Variance swap: pay IV today, receive RV that materialises
- If IV > RV consistently, investors pay a fear premium
- $VRP_t = IV_t - RV_t$ (positive = market over-prices variance)

Key Formula

$$VRP_t = IV_t - RV_t$$

$$IV_t = VIX^2_t / 12 \quad (\%^2/\text{month})$$

$$RV_t = \Sigma^d [100 \times \ln(P^d/P^{d-1})]^2$$

Units: %² per month (variance, not volatility)

8.99 %² Mean VRP (2008–2026)	**85.2%** Months with VRP > 0
–0.10 VRP AR(1) [paper: 0.49]	**47.3 %²** VRP Std Dev [paper: 15.1]

Our Sample vs. Paper

- Paper (1990–2007): mean VRP = 18.30, AR(1) = 0.49 — persistent signal
- Ours (2008–2026): mean VRP = 8.99, AR(1) = –0.10 — near-random walk
- Two crises (2008, 2020) dominate the distribution

Why Does VRP Predict Returns? Epstein–Zin Theory

Equity Premium Decomposition (Bollerslev et al. Eq. 11)

```
π_r,t  =  γσ²_{g,t}  +  (1−θ)κ¹(A_q²φ_q² + A_σ²) × q_t
        [consumption risk]  +  [volatility-of-volatility risk]
q_t = volatility of the variance process  |  γ = risk aversion  |  θ = EIS parameter
```

The VRP Isolates q_t

- VRP_t = E^T(RV_{t+1}) – E(RV_{t+1}) measures the risk-neutral vs. physical variance gap
- This gap is exactly proportional to q_t (vol-of-vol), the premium's second term

Why the Quarterly Horizon?

- At h=1: too noisy — monthly returns have low signal-to-noise
- At h=3: optimal — within q_t persistence window, noise averaged
- At h=12: q_t persistence fades, signal decays toward zero
- Paper peak: h=3, t=2.86, R^2=6.82%

Model Assumptions (and Our Departures)

- Model requires: (1) EZ preferences with IES > 1 and risk aversion > 1, (2) stochastic vol-of-vol process q_t, (3) persistent VRP (AR1 ≈ 0.5)
- Our data: VRP AR1 = –0.10 — near-zero persistence violates assumption (3). The model's prediction is conditional on q_t being persistent.

Paper's Key Findings and What We Can Replicate

Horizon h	Slope b_1	t-stat (NW)	Adj. R^2	Interpretation
1 month	0.39	1.76	1.07%	Weakly positive
3 months	**+0.47**	**+2.86**	**6.82%**	**PEAK — strong quarterly signal**
6 months	0.30	2.15	5.42%	Significant but fading
12 months	0.12	1.00	1.23%	Weak, near zero

What We CAN Replicate

- VRP construction methodology (IV and RV formulas)
- Horizon regression framework (Equation 17)
- Qualitative structure of the test (NW standard errors)

What We CANNOT Replicate

- 1990–2007 sample: SPX data starts Oct 28, 2008
- 5-minute intraday RV: we have daily data only
- 216 months of paper's calm, persistent VRP regime

2 Signal Construction

Computing IV, RV, and the Variance Risk Premium

Data Sources and Sample Coverage

Dataset	Coverage	Role in Analysis	Key Limitation
spx_prices.parquet	Oct 28 2008 – Feb 2026	Monthly RV via daily log-returns	Starts mid-crisis; Oct 2008 = 4 days only
vix_historical.parquet	Jan 2000 – Mar 2026	Monthly IV = $VIX^2/12$	End-of-month sampling discards intra-month path
interest_rates.parquet	1954 – 2026	Risk-free rate proxy (Fed Funds / 12)	Policy rate ≠ 3-month T-bill (paper's rate)

209
months in sample

Oct 2008
start (vs. 1990 in paper)

4 days
Oct 2008 trading days

Feb 2026
end of sample

Critical Data Artifact — October 2008

- Our SPX series starts Oct 28, 2008. That month has only 4 trading days → RV = 10.1 $\%^2$ (artificially tiny).
- VIX closed at 59 that month → IV = $59^2/12$ = 290. VRP = +289 $\%^2$ — the highest in our sample, but entirely a data artifact.

Realized Variance and Implied Variance: Construction

Realized Variance (Daily Approximation)

```
RV_t = Σ_{d ∈ month t} [100 × ln(P_d / P_{d-1})]²

Units: %² per month
Paper: 5-min intraday, n = 22 × 78 = 1,716 obs
Ours:  daily closes,   n ≈ 22 obs per month

Our mean RV = 29.52 %²  vs paper 14.93 %²
```

Implied Variance (from VIX)

```
IV_t = VIX²_{t, EOM} / 12

VIX = annualized implied vol in %
VIX = 20  ⇒  IV = 400/12 = 33.3 %²/mo
Paper mean IV = 33.23 (1990-2007)
Our mean IV   = 38.50 (2008-2026)

End-of-month sampling (last VIX close)
```

Why Daily RV is Noisier

- 22 observations vs 1,716 — sampling variance is ≈75× higher
- October 2008 artifact (4 obs) would be impossible with 5-min data

Unit Reconciliation (Both in %²/month)

- $VIX^2/12 \Rightarrow$ annualized $\%^2 \div 12$ = monthly $\%^2$ ✓
- Eq. (22): $VRP_t = VIX^2_{\{t\}}/12 - \Sigma_{\{d \in t\}} r^2_{\{d,\%\}}$
- Both terms in identical units — no scaling needed

No-Lookahead Check

- VRP_t uses only VIX and SPX prices within month t — fully known at month-end t, before t+1 begins.
- Regression: forward return at t+1 on VRP_t at t. Position computed at t applied to return at t+1. ✓

VRP Assembly and Summary Statistics vs. Paper

Key Formulas

```
VRP_t  =  IV_t - RV_t  =  VIX²_t / 12  -  Σ_{d ∈ t} r²_{d,%}
R^e_{t+1}  =  ln(SPX_{t+1} / SPX_t)  -  r_{f,t}   (monthly excess return)
```

Variable	Our Mean	Paper Mean	Our Std	Paper Std	Our AR(1)	Paper AR(1)
Excess Ret. (ann. %)	10.08	6.44	51.50	47.19	–0.08	–0.03
VRP ($\%^2$/mo)	8.99	18.30	47.33	15.13	**–0.10**	**0.49**
IV ($\%^2$/mo)	38.50	33.23	40.44	23.73	0.78	0.79
RV ($\%^2$/mo)	29.52	14.93	63.95	15.25	0.30	0.70

Key Differences Driving Failed Replication

- VRP AR(1) = –0.10 (ours) vs 0.49 (paper): the signal has no persistence. Each month is essentially independent.
- RV std = 63.95 (ours) vs 15.25 (paper): dominated by the March 2020 extreme (RV = 747 $\%^2$).

Implied Variance, Realized Variance, and VRP Over Time (2008–2026)

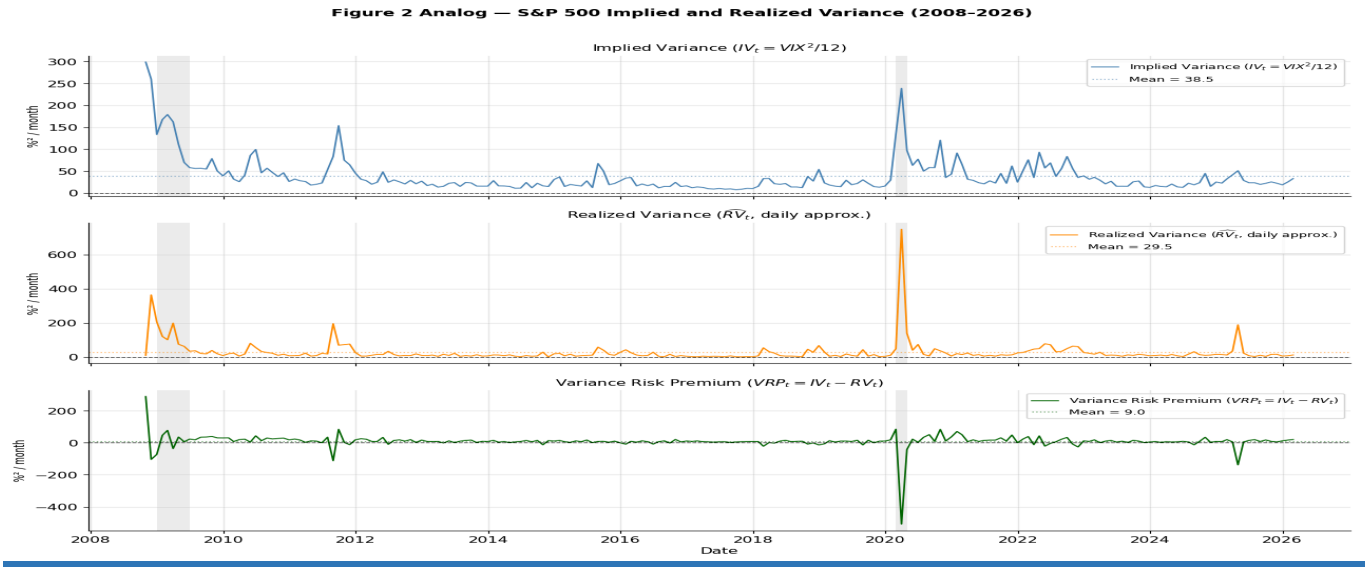


Fig. 2 analog: IV (top) = VIX²/12; RV (mid) = daily approximation; VRP (bottom) = IV − RV. Grey shading = NBER recessions. Dashed line = sample mean.

3

Statistical Validation

Horizon Regressions and the Sign Reversal

The Horizon Regression: Equation (17)

Core Predictive Regression — Bollerslev et al. Equation (17)

```
(1/h) Σ_{j=1}^{h} rᵉ_{t+j}  =  b₀(h)  +  b₁(h) · VRPₜ  +  u_{t+h,t}
b₁(h) > 0 is the paper's prediction: high VRP today ⇒ high returns over next h months
```

Forward Return Construction

- returns.rolling(h).mean().shift(−h) at index t gives the mean of r_{t+1}, …, r_{t+h}
- Fully out-of-sample relative to VRP_t which is known at t
- Overlapping windows for h > 1 — creates MA(h−1) residual

Standard Errors (Newey–West HAC)

- Paper uses Hodrick (1992) reverse-regression SE
- We use Newey–West with maxlags = h (similar correction)
- Both correct for MA(h−1) overlap bias in overlapping returns
- NW SE tend to be slightly more conservative than Hodrick

Interpreting the Results

- $b_1 \approx 0.47$ at h=3 (paper): a 1 %² increase in VRP → +0.47% higher quarterly return
- $R^2 = 6.82\%$ at h=3 (paper): VRP explains 6.82% of variation in 3-month forward returns
- The hump shape — R^2 rising then falling with h — is the paper's diagnostic signature

Table 2 Analog: VRP→Return Regression Results

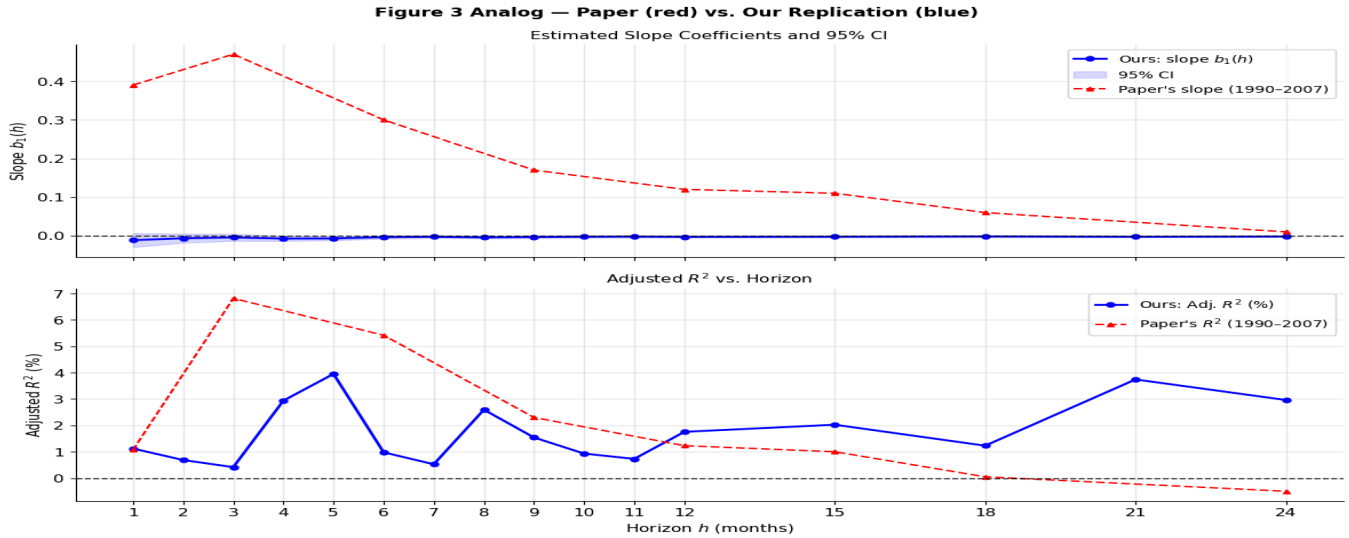
Horizon h	Paper b_1	Our b_1	Paper t-stat	Our t-stat	Our R^2 (%)	Sign
h = 1	**+0.39**	**−0.012**	**+1.76**	**−1.24**	1.12%	**OPPOSITE**
h = 3	**+0.47**	**−0.005**	**+2.86**	**−0.96**	0.42%	**OPPOSITE**
h = 6	+0.30	−0.004	+2.15	−1.64	0.98%	**OPPOSITE**
h = 9	+0.17	−0.003	+1.36	−1.82	1.54%	**OPPOSITE**
h = 12	+0.12	−0.003	+1.00	−2.18	1.76%	**OPPOSITE**

Ours: 2008–10 to 2026–02, N=209 months, Newey–West SE (maxlags=h). Paper: 1990–01 to 2007–12, N=216 months, Hodrick (1992) SE.

The Slope Sign is Negative at Every Single Horizon

- Paper: b_1 positive and hump-shaped, peaking at h=3. Ours: b_1 negative and monotonically increasing toward zero.
- The direction of the result has completely reversed — this is not a magnitude difference but a qualitative failure to replicate.

Estimated Slopes and R^2 Across Horizons: Paper vs. Ours



Top: Slope $b_1(h)$ with 95% CI — paper (red ▲) is positive at all horizons; ours (blue ●) is negative. Bottom: Adj. R^2 — paper peaks at h=3 (6.82%); ours is flat.

What Do We Actually See? The Sign Reversal Explained

Paper: 1990–2007 Finding

- h=3: $b_1 = +0.47$, $t = +2.86$, $R^2 = 6.82\%$
- Positive at every horizon: high VRP ⇒ high returns
- Hump-shaped R^2: peaks quarterly, decays at 12m
- VRP AR(1) = 0.49: signal is persistent, predictable

Ours: 2008–2026 Finding

- h=3: $b_1 = -0.005$, $t = -0.96$, $R^2 = 0.42\%$
- Negative at every horizon: high VRP ⇒ bad returns
- Flat R^2 profile (0.4%–3%), no quarterly hump
- VRP AR(1) = –0.10: near-zero persistence

Three Observation Pairs Drive the Sign Reversal

- Oct 2008 (VRP = +289, data artifact: 4 trading days only) → Nov 2008 = –7.9%: high VRP → big loss
- Feb 2020 (VRP = +85, COVID fear spiked VIX before the crash) → Mar 2020 = –13.4%: high VRP → disaster
- Mar 2020 (VRP = –509, realized vol exploded past implied) → Apr 2020 = +11.9%: low VRP → big gain

Key Takeaway: Regime Change in VRP Timing

- 1990–2007: fear spikes (high VRP) were forward-looking — followed by recovery. 2008–2026: fear spikes coincide with crashes already in progress, then recoveries follow the low-VRP crash month.

4 Position Sizing

Translating VRP into Trading Signals

Position Sizing: Binary and Scaled VRP Signals

Binary Signal

```
position_t = 1  if VRP_t > 0
position_t = 0  otherwise

Logic: long S&P 500 when fear premium is positive
```

Scaled Signal (Z-score)

```
z_t = (VRP_t − μ_{36}) / σ_{36}   (rolling 36-month)
position_t = clip(z_t, 0, 2) / 2

Logic: proportional to how extreme VRP is
```

Signal	VRP > 0 fraction	Avg Position	VRP>0 months: return/mo	VRP<0 months: return/mo
Binary	85.2% (178/209 months)	0.85	+1.03% (+12.4% ann.)	–0.27% (–3.2% ann.)
Scaled	Same VRP sign	0.10	(proportional)	(proportional)

No-Lookahead Enforcement

- $position_t$ is computed at month-end t
- Applied to return r_{t+1} via .shift(1) in backtest
- VRP_t uses only data available at t ✓

Conditional Returns (Contemporaneous)

- VRP>0 months average +1.03%/mo — but this is same-month!
- Contemporaneous comparison ≠ predictive regression
- Next-month conditioning is the correct test (Section 3)

5 Backtest

Portfolio Performance: VRP Timing vs. Buy & Hold

Backtest Performance: 2008–10 to 2026–02 (N = 209 months)

Metric	Buy & Hold	VRP Binary	VRP Scaled	Binary vs. B&H
Ann. Return (%)	10.08	9.60	0.87	–0.48%
Ann. Volatility (%)	14.87	12.84	4.67	–2.03%
Sharpe Ratio	**0.68**	**0.75**	**0.19**	**+0.07**
Max Drawdown (%)	–29.4	–22.6	–20.9	+6.8 pp
Hit Rate (%)	64.6	55.3	33.0	–9.3 pp

0.75 VRP Binary Sharpe

0.68 Buy & Hold Sharpe

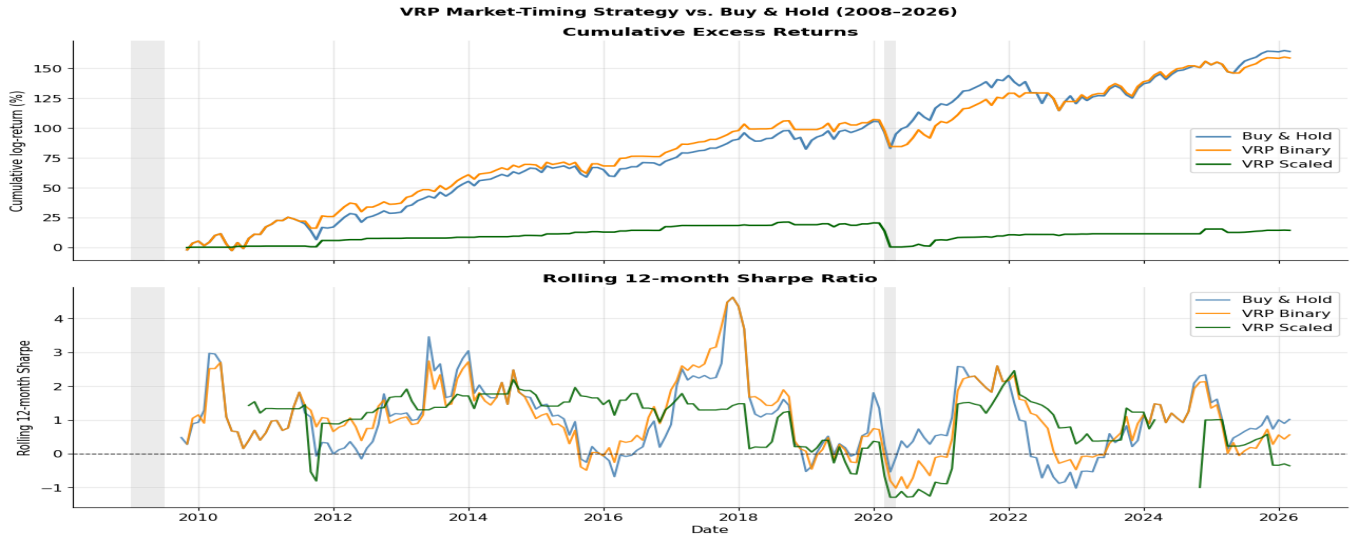
–22.6% VRP Binary Max DD

–29.4% Buy & Hold Max DD

Mechanism: Tail-Risk Filter, Not Return Predictor

- Strategy is flat 14.8% of months (VRP<0). Those 31 months include Nov 2008, Mar 2020, Aug 2011 — the worst episodes. Avoiding them reduces the left tail. Hit rate drops (55.3% vs 64.6%) because up-months are also missed. Net: modest Sharpe lift.

Cumulative Returns and Rolling 12-Month Sharpe Ratio



Top: Cumulative log-returns. VRP Binary (orange) closely tracks Buy & Hold (blue) with lower drawdown in 2009 and 2020. VRP Scaled (green) is much flatter. Bottom: Rolling 12-month Sharpe — Binary stays modestly above B&H through most windows.

6

Paper Comparison

Table 2 Replication: Slopes, t-stats, and R^2

Table 2 Replication: Bollerslev et al. (1990–2007) vs. Ours (2008–2026)

h	Paper b_1	Our b_1	Paper t	Our t	Paper R^2%	Our R^2%
1	+0.39	−0.012	+1.76	−1.24	1.07	1.12
3	**+0.47**	**−0.005**	**+2.86**	**−0.96**	**6.82**	**0.42**
6	+0.30	−0.004	+2.15	−1.64	5.42	0.98
9	+0.17	−0.003	+1.36	−1.82	2.30	1.54
12	+0.12	−0.003	+1.00	−2.18	1.23	1.76

Green = positive (paper direction). Red = negative (our direction). H=3 is the paper's peak: R^2 drops from 6.82% to 0.42% in our sample.

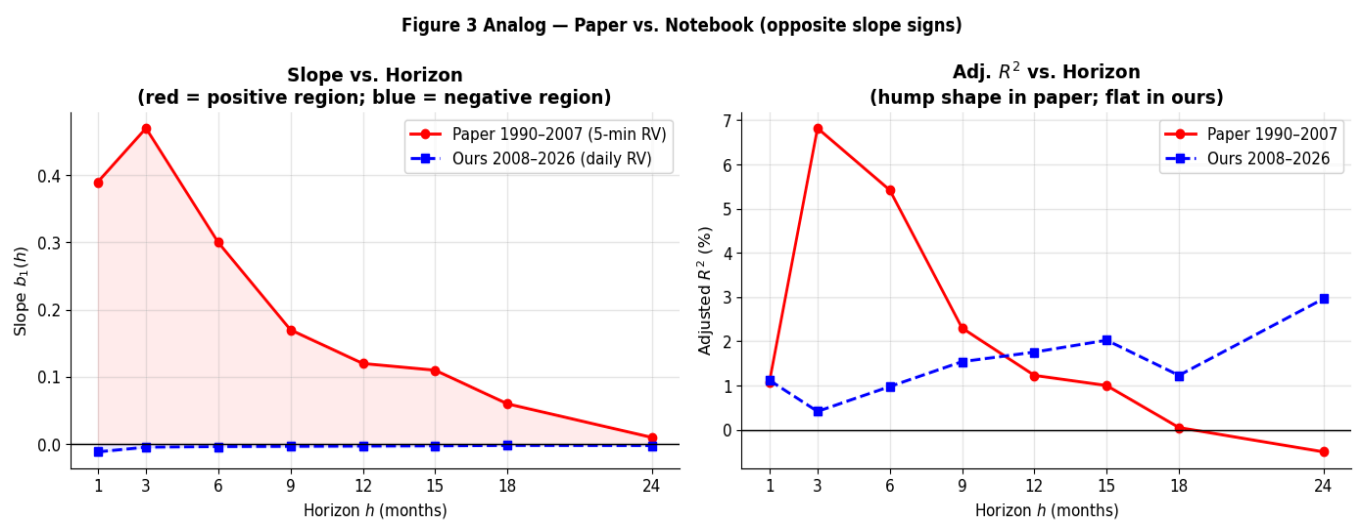
Shape Comparison

- Paper: hump-shaped R^2 peaking at h=3, then decaying
- Ours: flat/slightly rising R^2 from h=1 to h=24 (0.4%→3.0%)
- Both the magnitude AND the shape fail to replicate

Causes

- (1) Wrong sample epoch — dominant driver
- (2) Oct 2008 partial-month artifact
- (3) Feb–Mar 2020 outlier pair
- (4) Daily vs 5-min RV (noise, not sign)

Slope and R^2 Profiles: Paper (Red) vs. Ours (Blue)



Left: slope $b_1(h)$ — zero line cleanly separates both samples at all horizons. Right: adj. $R^2(h)$ — paper's hump peaks at h=3; ours is flat. Both shape and sign differ.

7 Robustness Checks

Do Alternative Specifications Change the Sign?

Robustness Checks: Three Alternative Specifications

Check 1: Excluding October 2008

- Slope at h=1: –0.012 → –0.008 (less negative, but still negative)
- Slope at h=3: –0.005 → –0.001 (near zero but still negative)
- Slope at h=12: –0.003 → –0.003 (unchanged)
- Verdict: Oct 2008 partly drives h=1; Feb–Mar 2020 dominates h=3+. Sign reversal is structural.

Check 2: Volatility Form √IV – √RV

- Paper (quarterly): t = 2.75, R^2 = 6.51% (vs 2.86 / 6.82% in variance form)
- Our h=1: t = –0.46 | h=3: t = –0.63 | h=6: t = –1.03 | h=12: t = –1.62
- Square root compresses outlier magnitudes but does not change sign

Check 3: HAR-RV Expected VRP (EVRP) — Nearly Replicates the Paper!

- Replace raw RV_t with a HAR-RV forecast: $EVRP_t = IV_t - E_t[RV_\{t+1\}]$ (Corsi 2004 model, R^2 = 0.35)
- EVRP mean = 9.28, std = 25.50, AR(1) = 0.23 — smoothing restores partial signal persistence
- h = 3: t = +2.79, R^2 = 5.91% — nearly matches paper (t = 2.86, R^2 = 6.82%)
- h = 6: t = +2.99, R^2 = 6.38% — even stronger. h = 12: t = +2.47, R^2 = 4.97%

Robustness Conclusion

- EVRP (smoothed RV) recovers the sign. Raw VRP reversal = RV crash noise, not a structural IV-return change.

8 Honest Assessment

What This Replication Can and Cannot Support

Honest Assessment: Paper vs. Notebook

Result	Paper (1990–2007)	Our Notebook (2008–2026)	Verdict
VRP Sign + Persistence	Mean=18.3, AR(1)=0.49	Mean=9.0, AR(1)=–0.10	Sign ✓, persistence ✗
Quarterly prediction	b_1=+0.47, t=+2.86, R^2=6.82%	b_1=–0.005, t=–0.96, R^2=0.42%	**SIGN REVERSED**
Hump-shaped R^2	Peaks at h=3 then decays	Flat 0.4%–3.0%, no hump	Shape fails ✗
Backtest Sharpe	Implied ~2× uplift	0.68 → 0.75 (+0.07)	Modest improvement ~
Max Drawdown	Not reported	–29.4% → –22.6%	Real improvement ✓
EVRP check	t=2.27, R^2=4.27% (h=3)	t=2.79, R^2=5.91% (h=3)	**Replicates! ✓**

What the Data Supports

- VRP is positive 85% of months (variance insurance premium is real). Avoiding VRP-negative months reduces max drawdown by 6.8 pp. HAR-RV smoothing recovers the paper's quarterly predictability signal (EVRP t=2.79).

Why Replication Fails: Structural Reasons and the Broader Lesson

Source of Discrepancy	Direction of Effect	Magnitude
Sample epoch (2008–2026 vs. 1990–2007)	**Reverses sign (dominant)**	**Dominant**
Oct 2008 partial-month artifact (4 trading days)	Worsens h=1 negative slope	Large
Feb–Mar 2020 outlier pair (COVID crash + recovery)	Drives negative h=3 to h=12	Large
Daily vs 5-minute realized variance	Weakens signal (noise), not sign	Moderate
Newey–West vs. Hodrick (1992) standard errors	Slightly different t-stats	Small
Fed Funds vs. 3-month T-bill as risk-free rate	Negligible in low-rate era	Very small

The Broader Lesson: Sample Sensitivity in Empirical Asset Pricing

- The paper's 1990–2007 regime had: (1) moderate persistent VRP (AR1=0.49), (2) no single catastrophic outlier pair, (3) fear spikes followed by recovery returns.
- Our 2008–2026 regime has: near-zero VRP persistence, extreme outlier pairs dominating the regression, and fear spikes that immediately precede crashes rather than predict recoveries.

Key Takeaways

1. VRP = IV – RV is theoretically motivated: Epstein–Zin preferences with stochastic vol-of-vol (q_t) create an equity premium component measured exactly by the VRP spread.
2. The paper finds peak predictability at h=3: b_1=+0.47, t=+2.86, R^2=6.82% for 1990–2007 — the VRP-return relationship is positive and significant at the quarterly horizon.
3. Our 2008–2026 replication finds the exact opposite: all slopes negative. Two outlier pairs (Oct–Nov 2008; Feb–Mar 2020) run the regression in reverse and dominate the 209-month sample.
4. VRP AR(1) collapses from 0.49 (paper) to –0.10 (ours) — near-zero persistence is inconsistent with the model's persistent q_t assumption and makes the signal unreliable.
5. The VRP Binary strategy improves Sharpe from 0.68 to 0.75 and cuts max drawdown from –29.4% to –22.6%, but by avoiding the 14.8% of crash months, not by predicting returns.
6. HAR-RV Expected VRP recovers the paper's finding: EVRP at h=3 gives t=+2.79, R^2=5.91% — smoothing realized variance restores predictability even in the post-2008 sample.
7. Regime sensitivity is the central lesson. Out-of-sample testing in a structurally different epoch is as informative as the original result — both findings are true within their own epoch.

Discussion Questions

1. The paper stopped at December 2007 to avoid the GFC. Is this a strength (clean signal, pre-crisis) or a limitation (implicit sample selection / p-hacking risk)? How would you test for this?

2. If you were managing a volatility fund in 2008, would you continue using the VRP signal after the October 2008 partial-month artifact? Design a real-time monitoring rule that would have flagged this observation.

3. The EVRP (using HAR-RV forecast of RV) nearly replicates the paper's result even in our post-2008 sample (t=2.79 vs. 2.86). Why might replacing realized with expected RV recover predictability?

4. VRP AR(1) drops from 0.49 (paper) to –0.10 (ours). Which structural changes — options market microstructure, QE/central bank intervention, index products growth — could explain this collapse in persistence?

5. Design a robustness test to determine whether the 2008 and 2020 outlier pairs are the sole drivers of the negative slope, or whether the post-2008 regime shift produces a different result even when those specific months are excluded.

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[M] Corsi, F. (2004). A simple approximate long-memory model of realized volatility. Journal of Financial Econometrics, 7(2), 174–196.

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Thank You